Overall Project Summary and Analytical Reasoning

This project used transfer learning with ResNet50 to classify CIFAR-10 images into 10 categories such as airplanes, cats, dogs, ships, and trucks. CIFAR-10 is a challenging dataset because the images are very small (32×32 pixels), contain limited visual detail, and include several visually similar classes. The overall goal of this project was not only to build a working CNN model, but also to understand the reasoning behind preprocessing, architectural choices, training strategies, and performance evaluation.

Preprocessing Choices

Several preprocessing steps were applied before training. First, pixel values were normalized and passed through preprocess_input() because ResNet50 was originally trained on ImageNet using this preprocessing format. Matching the original preprocessing helps the pretrained filters interpret CIFAR-10 images correctly and improves transfer learning performance.

The dataset was also divided into training, validation, and final test sets. The validation set was used during training to monitor generalization performance and detect overfitting, while the test set remained completely untouched until the final evaluation to ensure unbiased results.

Data augmentation techniques such as horizontal flipping, rotation, and zooming were added to artificially increase dataset variability. This helps reduce overfitting and improves the model's ability to generalize to unseen images.

Training the Custom Head vs Fine-Tuning

The project was divided into two main stages: frozen-base training and fine-tuning.

During the first stage, the ResNet50 base model was frozen, meaning its pretrained weights remained unchanged. Only the custom Dense classification head was trained. This allowed the model to reuse previously learned visual features such as edges, textures, and shapes while learning how to classify CIFAR-10 images. Freezing the base model also reduced training time and lowered the risk of overfitting during the initial stage.

In the second stage, deeper ResNet50 layers were unfrozen for fine-tuning. This allowed the pretrained network to adapt more specifically to CIFAR-10 features. A smaller learning rate was used during fine-tuning to avoid damaging the pretrained weights with large updates.

Observations During Training

The training curves showed that the frozen ResNet50 model learned stable and useful visual patterns. Training accuracy steadily increased while validation accuracy improved and later stabilized around 64–65%. Validation loss remained relatively stable, indicating reasonable generalization performance.

During fine-tuning, training accuracy continued increasing while training loss decreased further. However, validation accuracy improved only slightly and then plateaued, while validation loss remained almost stable. This indicated mild overfitting, meaning the model became more specialized to the training data without significantly improving performance on unseen validation images.

The confusion matrix provided additional insight into model behavior. The model performed well on visually distinctive classes such as frogs, ships, airplanes, and automobiles. However, it struggled more with visually similar categories such as cats vs dogs and deer vs horses. These confusions are expected because CIFAR-10 images are small and contain limited visual detail.

Constraints and Limitations

Several practical limitations affected the project. CIFAR-10 images are low-resolution, making fine visual distinctions difficult. In addition, only a subset of the full dataset was used to reduce runtime and memory usage in Google Colab. Hardware and runtime limitations also restricted experimentation with larger architectures, longer training schedules, and deeper fine-tuning.

Another important limitation was balancing model complexity and generalization. Increasing the number of trainable layers improved learning capacity but also increased the risk of overfitting.

Final Conclusion

Overall, this project successfully demonstrated how transfer learning can be applied to image classification using CNNs and ResNet50. The preprocessing pipeline, frozen-base training stage, and fine-tuning stage each served different purposes and highlighted important machine learning concepts such as feature reuse, generalization, and overfitting.

The final model achieved approximately **65% test accuracy**, demonstrating that transfer learning with ResNet50 can effectively learn meaningful visual patterns even on small low-resolution CIFAR-10 images. While the model did not achieve state-of-the-art performance, the project provided valuable insight into how deep learning models are trained, evaluated, interpreted, and improved in practice.

Main Ways to Improve Model Performance

Several improvements could further enhance the CIFAR-10 ResNet50 model and improve generalization performance:

1. Use the full CIFAR-10 dataset or a larger subset
2. Improve data augmentation techniques
3. Fine-tune more pretrained ResNet50 layers gradually

4. Tune the learning rate and training schedule
5. Optimize dropout and regularization settings
6. Improve the custom Dense classification head
7. Train longer while using EarlyStopping
8. Use stronger validation strategies
9. Increase input image resolution
10. Try alternative pretrained CNNs such as EfficientNet or DenseNet
11. Perform systematic hyperparameter tuning
12. Apply stronger overfitting reduction techniques
13. Analyze confusion matrix errors to refine preprocessing and architecture choices